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What Are You Searching For? A Remote Keylogging Attack on Search Engine Autocomplete

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Many search engines have an autocomplete feature that presents a list of suggested queries to the user as they type. Autocomplete induces network traffic from the client upon changes to the query in a web page. We describe a remote keylogging attack on search engine autocomplete. The attack integrates information leaked by three independent sources: the timing of keystrokes manifested in packet inter-arrival times, percent-encoded Space characters in a URL, and the static Huffman code used in HTTP2 header compression. While each source is a relatively weak predictor in its own right, combined, and by leveraging the relatively low entropy of English language, up to 15% of search queries are identified among a list of 50 hypothesis queries generated from a dictionary with over 12k words. The attack succeeds despite network traffic being encrypted. We demonstrate the attack on two popular search engines and discuss some countermeasures to mitigate attack success.

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